

B5: General Relativity — 2024 Model Solutions

Trinity Term 2024 (Paper A16729W1)

Question 1 — Perfect-fluid conservation, modified Einstein equations

Problem. (a) Find a current j^α obeying $j^\alpha{}_{;\alpha} = U^\alpha P_{,\alpha}$ from $T^{\alpha\beta}{}_{;\alpha} = 0$ for the perfect-fluid stress-energy. (b) Show that pressureless flow is geodesic. (c) The set $R_{\alpha\beta} - (1/n)Rg_{\alpha\beta} = 0$: count equations, find the special n , show R is constant. (d) Show that a trace-modified Einstein equation reduces to standard Einstein with cosmological constant. (e) Find Λ for $R_{\alpha\beta\mu\nu} = (K/12)(g_{\alpha\mu}g_{\beta\nu} - g_{\alpha\nu}g_{\beta\mu})$ in vacuum.

(a) Current from $T^{\alpha\beta}{}_{;\alpha} = 0$

Take the divergence of $T^{\alpha\beta}$, using $g^{\alpha\beta}{}_{;\alpha} = 0$:

$$T^{\alpha\beta}{}_{;\alpha} = (Pg^{\alpha\beta})_{;\alpha} + [(\rho + P/c^2)U^\alpha U^\beta]_{;\alpha}.$$

Pull the metric outside the derivative:

$$T^{\alpha\beta}{}_{;\alpha} = g^{\alpha\beta}P_{,\alpha} + \frac{1}{c^2}[(\rho c^2 + P)U^\alpha U^\beta]_{;\alpha}.$$

Set the divergence to zero:

$$g^{\alpha\beta}P_{,\alpha} + \frac{1}{c^2}[(\rho c^2 + P)U^\alpha U^\beta]_{;\alpha} = 0. \quad (1)$$

Define the candidate current:

$$j^\alpha \equiv (\rho c^2 + P)U^\alpha.$$

Insert into (??):

$$g^{\alpha\beta}P_{,\alpha} + \frac{1}{c^2}(j^\alpha U^\beta)_{;\alpha} = 0.$$

Apply the Leibniz rule to the product:

$$\frac{1}{c^2}(j^\alpha U^\beta)_{;\alpha} = \frac{1}{c^2}j^\alpha{}_{;\alpha}U^\beta + \frac{1}{c^2}j^\alpha U^\beta{}_{;\alpha}.$$

Rearrange the equation:

$$\frac{1}{c^2}j^\alpha{}_{;\alpha}U^\beta + \frac{1}{c^2}j^\alpha U^\beta{}_{;\alpha} = -g^{\alpha\beta}P_{,\alpha} = -P^{,\beta}.$$

Contract with U_β :

$$\frac{1}{c^2}j^\alpha{}_{;\alpha}(U_\beta U^\beta) + \frac{1}{c^2}j^\alpha U_\beta U^\beta{}_{;\alpha} = -U_\beta P^{,\beta}.$$

Use $U_\beta U^\beta = -c^2$ on the first term:

$$-j^\alpha{}_{;\alpha} + \frac{1}{c^2}j^\alpha U_\beta U^\beta{}_{;\alpha} = -U^\alpha P_{,\alpha}.$$

Apply $U_\beta U^\beta{}_{;\alpha} = \frac{1}{2}(U_\beta U^\beta)_{;\alpha} = 0$:

$$-j^\alpha{}_{;\alpha} = -U^\alpha P_{,\alpha}.$$

Multiply through by -1 :

$$j^\alpha{}_{;\alpha} = U^\alpha P_{,\alpha}.$$

$j^\alpha = (\rho c^2 + P)U^\alpha, \quad j^\alpha{}_{;\alpha} = U^\alpha P_{,\alpha}.$

(2)

(b) Pressureless flow is geodesic

Set $P = 0$ in the stress-energy tensor:

$$T^{\alpha\beta} = \rho U^\alpha U^\beta.$$

Take the divergence and apply Leibniz:

$$T^{\alpha\beta}_{;\alpha} = (\rho U^\alpha)_{;\alpha} U^\beta + \rho U^\alpha U^\beta_{;\alpha} = 0.$$

Contract with U_β :

$$(\rho U^\alpha)_{;\alpha} (U_\beta U^\beta) + \rho U^\alpha U_\beta U^\beta_{;\alpha} = 0.$$

Use $U_\beta U^\beta = -c^2$ and $U_\beta U^\beta_{;\alpha} = 0$:

$$-c^2 (\rho U^\alpha)_{;\alpha} = 0.$$

Divide by $-c^2$ to obtain rest-mass conservation:

$$(\rho U^\alpha)_{;\alpha} = 0.$$

Substitute back into the divergence:

$$0 \cdot U^\beta + \rho U^\alpha U^\beta_{;\alpha} = 0.$$

Divide by ρ :

$$U^\alpha U^\beta_{;\alpha} = 0.$$

$$\boxed{U^\alpha U^\beta_{;\alpha} = 0.} \tag{3}$$

(c) The set $R_{\alpha\beta} - (1/n)Rg_{\alpha\beta} = 0$

For $n = 2$ the equation reads $R_{\alpha\beta} - \frac{1}{2}Rg_{\alpha\beta} = 0$ (vacuum Einstein). The tensor $R_{\alpha\beta} - (1/n)Rg_{\alpha\beta}$ is symmetric; in 4D it carries:

$$\frac{4(4+1)}{2} = 10 \text{ independent components.}$$

Take the trace $g^{\alpha\beta}(\cdot)$ of the equation:

$$g^{\alpha\beta} R_{\alpha\beta} - \frac{1}{n} R g^{\alpha\beta} g_{\alpha\beta} = 0.$$

Use $g^{\alpha\beta} R_{\alpha\beta} = R$ and $g^{\alpha\beta} g_{\alpha\beta} = 4$:

$$R - \frac{4}{n} R = 0.$$

Factor R :

$$\left(1 - \frac{4}{n}\right) R = 0.$$

The trace becomes automatic for $n = 4$, leaving one fewer independent equation:

$$\boxed{n = 4 \Rightarrow 9 \text{ independent equations.}} \tag{4}$$

Raise an index on the field equation:

$$R^\beta{}_\alpha = \frac{1}{n} R \delta^\beta{}_\alpha.$$

Take the covariant divergence:

$$R^\beta{}_{\alpha;\beta} = \frac{1}{n} R_{,\beta} \delta^\beta{}_\alpha = \frac{1}{n} R_{,\alpha}.$$

Apply the contracted Bianchi identity $R_{,\alpha} = 2R^\beta_{\alpha;\beta}$:

$$R_{,\alpha} = \frac{2}{n} R_{,\alpha}.$$

Rearrange:

$$\left(1 - \frac{2}{n}\right) R_{,\alpha} = 0.$$

Substitute $n = 4$:

$$\frac{1}{2} R_{,\alpha} = 0.$$

Hence:

$$\boxed{R = \text{const.}}$$

(d) Trace-free modified Einstein $\Rightarrow$ standard Einstein $+\Lambda$

State the modified equation:

$$R_{\alpha\beta} - \frac{1}{4} R g_{\alpha\beta} = -\frac{8\pi G}{c^4} (T_{\alpha\beta} - \frac{1}{4} T g_{\alpha\beta}). \quad (5)$$

Take the covariant divergence of both sides:

$$\nabla^\beta R_{\alpha\beta} - \frac{1}{4} R_{,\alpha} = -\frac{8\pi G}{c^4} (\nabla^\beta T_{\alpha\beta} - \frac{1}{4} T_{,\alpha}).$$

Apply $\nabla^\beta R_{\alpha\beta} = \frac{1}{2} R_{,\alpha}$ and $\nabla^\beta T_{\alpha\beta} = 0$:

$$\frac{1}{2} R_{,\alpha} - \frac{1}{4} R_{,\alpha} = -\frac{8\pi G}{c^4} (0 - \frac{1}{4} T_{,\alpha}).$$

Combine like terms:

$$\frac{1}{4} R_{,\alpha} = \frac{2\pi G}{c^4} T_{,\alpha}.$$

Multiply through by 4:

$$R_{,\alpha} = \frac{8\pi G}{c^4} T_{,\alpha}.$$

Integrate, calling the integration constant 4Λ :

$$R = \frac{8\pi G}{c^4} T + 4\Lambda. \quad (6)$$

Substitute (??) into (??):

$$R_{\alpha\beta} - \frac{1}{4} \left(\frac{8\pi G}{c^4} T + 4\Lambda \right) g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta} + \frac{2\pi G}{c^4} T g_{\alpha\beta}.$$

Expand the bracket on the LHS:

$$R_{\alpha\beta} - \frac{2\pi G}{c^4} T g_{\alpha\beta} - \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta} + \frac{2\pi G}{c^4} T g_{\alpha\beta}.$$

Move the $T g_{\alpha\beta}$ terms to one side:

$$R_{\alpha\beta} = \Lambda g_{\alpha\beta} + \frac{4\pi G}{c^4} T g_{\alpha\beta} - \frac{8\pi G}{c^4} T_{\alpha\beta}.$$

Form $\frac{1}{2} R g_{\alpha\beta}$ using (??):

$$\frac{1}{2} R g_{\alpha\beta} = \frac{4\pi G}{c^4} T g_{\alpha\beta} + 2\Lambda g_{\alpha\beta}.$$

Subtract from the previous expression for $R_{\alpha\beta}$:

$$R_{\alpha\beta} - \frac{1}{2} R g_{\alpha\beta} = \Lambda g_{\alpha\beta} - 2\Lambda g_{\alpha\beta} + \frac{4\pi G}{c^4} T g_{\alpha\beta} - \frac{4\pi G}{c^4} T g_{\alpha\beta} - \frac{8\pi G}{c^4} T_{\alpha\beta}.$$

Cancel the $T g_{\alpha\beta}$ terms and combine Λ terms:

$$R_{\alpha\beta} - \frac{1}{2} R g_{\alpha\beta} = -\Lambda g_{\alpha\beta} - \frac{8\pi G}{c^4} T_{\alpha\beta}.$$

Move $\Lambda g_{\alpha\beta}$ to the LHS:

$$\boxed{R_{\alpha\beta} - \frac{1}{2} R g_{\alpha\beta} - \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta}.} \quad (7)$$

This is the standard Einstein equation with cosmological constant Λ .

(e) Maximally symmetric vacuum: Λ from K

Use the paper's convention $R_{\alpha\beta} \equiv -R^\gamma_{\alpha\beta\gamma}$. Raise the first index of the given Riemann tensor with $g^{\gamma\sigma}$:

$$R^\gamma_{\alpha\beta\delta} = g^{\gamma\sigma} R_{\sigma\alpha\beta\delta} = \frac{K}{12} (\delta^\gamma_\beta g_{\alpha\delta} - \delta^\gamma_\delta g_{\alpha\beta}).$$

Contract on $\gamma = \delta$:

$$R^\gamma_{\alpha\beta\gamma} = \frac{K}{12} (\delta^\gamma_\beta g_{\alpha\gamma} - \delta^\gamma_\gamma g_{\alpha\beta}).$$

Use $\delta^\gamma_\beta g_{\alpha\gamma} = g_{\alpha\beta}$ and $\delta^\gamma_\gamma = 4$:

$$R^\gamma_{\alpha\beta\gamma} = \frac{K}{12} (g_{\alpha\beta} - 4g_{\alpha\beta}).$$

Combine the terms in the bracket:

$$R^\gamma_{\alpha\beta\gamma} = -\frac{K}{4} g_{\alpha\beta}.$$

Apply the Ricci sign convention:

$$R_{\alpha\beta} = -R^\gamma_{\alpha\beta\gamma} = \frac{K}{4} g_{\alpha\beta}.$$

Take the trace with $g^{\alpha\beta}$:

$$R = \frac{K}{4} g^{\alpha\beta} g_{\alpha\beta} = \frac{K}{4} \cdot 4 = K.$$

Insert into (??) with $T_{\alpha\beta} = 0$:

$$\frac{K}{4} g_{\alpha\beta} - \frac{1}{2} K g_{\alpha\beta} - \Lambda g_{\alpha\beta} = 0.$$

Collect the coefficients of $g_{\alpha\beta}$:

$$\left(\frac{K}{4} - \frac{K}{2} - \Lambda\right) g_{\alpha\beta} = 0.$$

Combine the K terms:

$$-\frac{K}{4} - \Lambda = 0.$$

Solve for Λ :

$$\boxed{\Lambda = -\frac{K}{4}}. \tag{8}$$

Question 2 — Reissner–Nordström geodesics

Problem. Charged BH metric, $A(r) = 1 - 2GM/(c^2 r) + GQ^2/(c^4 r^2)$. (a) Conserved k, h . (b) Effective-potential equation. (c) Particle from rest at infinity: turning point r_* . (d) Free fall from rest at r_+ when $r_Q > r_G$: $\dot{r}^2$ form, motion. (e) Proper time from r_+ to r_- .

(a) Conserved quantities along the geodesic

The metric is independent of t and ϕ . Write the Lagrangian on the equator ($\theta = \pi/2$, $\dot{\theta} = 0$):

$$2\mathcal{L} = -Ac^2\dot{t}^2 + \frac{\dot{r}^2}{A} + r^2\dot{\phi}^2.$$

Apply the Euler–Lagrange equation for t :

$$\frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \dot{t}} = \frac{\partial \mathcal{L}}{\partial t} = 0.$$

Evaluate the momentum:

$$\frac{\partial \mathcal{L}}{\partial \dot{t}} = -Ac^2\dot{t}.$$

The momentum is therefore conserved:

$$-Ac^2\dot{t} = \text{const.}$$

Absorb constants into the conserved energy parameter k :

$$\boxed{At = k.} \quad (9)$$

Apply the Euler–Lagrange equation for ϕ :

$$\frac{d}{d\tau}(r^2 \sin^2 \theta \dot{\phi}) = 0.$$

Set $\sin \theta = 1$ on the equator:

$$\frac{d}{d\tau}(r^2 \dot{\phi}) = 0.$$

Integrate to define the angular momentum h :

$$\boxed{r^2 \dot{\phi} = h.} \quad (10)$$

(b) Effective-potential equation

Apply the timelike normalisation on the equator:

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -Ac^2 \dot{t}^2 + \frac{\dot{r}^2}{A} + r^2 \dot{\phi}^2 = -c^2.$$

Substitute $\dot{t} = k/A$ and $\dot{\phi} = h/r^2$:

$$-Ac^2 \frac{k^2}{A^2} + \frac{\dot{r}^2}{A} + \frac{h^2}{r^2} = -c^2.$$

Simplify the first term:

$$-\frac{c^2 k^2}{A} + \frac{\dot{r}^2}{A} + \frac{h^2}{r^2} = -c^2.$$

Multiply both sides by A :

$$-c^2 k^2 + \dot{r}^2 + \frac{Ah^2}{r^2} = -Ac^2.$$

Solve for $\dot{r}^2$:

$$\dot{r}^2 = c^2 k^2 - c^2 A - \frac{Ah^2}{r^2}.$$

Divide by 2 and add $\frac{1}{2}c^2 A + Ah^2/(2r^2) - \frac{1}{2}c^2$ to both sides:

$$\frac{1}{2}\dot{r}^2 + \frac{1}{2}c^2(A - 1) + \frac{Ah^2}{2r^2} = \frac{1}{2}c^2(k^2 - 1).$$

Read off $V_{\text{eff}} = \frac{1}{2}c^2(A - 1) + Ah^2/(2r^2)$. Insert $A = 1 - r_G/r + r_Q^2/r^2$ and form $A - 1$:

$$A - 1 = -\frac{r_G}{r} + \frac{r_Q^2}{r^2}.$$

Multiply by $c^2/2$:

$$\frac{1}{2}c^2(A - 1) = -\frac{c^2 r_G}{2r} + \frac{c^2 r_Q^2}{2r^2}.$$

Expand $Ah^2/(2r^2)$ term by term:

$$\frac{Ah^2}{2r^2} = \frac{h^2}{2r^2} - \frac{h^2 r_G}{2r^3} + \frac{h^2 r_Q^2}{2r^4}.$$

Sum the two contributions:

$$V_{\text{eff}} = -\frac{c^2 r_G}{2r} + \frac{c^2 r_Q^2}{2r^2} + \frac{h^2}{2r^2} - \frac{h^2 r_G}{2r^3} + \frac{h^2 r_Q^2}{2r^4}.$$

Combine the $1/r^2$ terms:

$$\boxed{V_{\text{eff}}(r) = -\frac{c^2 r_G}{2r} + \frac{c^2 r_Q^2 + h^2}{2r^2} - \frac{h^2 r_G}{2r^3} + \frac{(hr_Q)^2}{2r^4}.} \quad (11)$$

(c) Stop point for a particle starting from rest at infinity

At rest at infinity: $h = 0$ and $A(\infty) = 1$. Apply the normalisation:

$$-c^2 = -Ac^2 \dot{t}^2 \Big|_{r \rightarrow \infty} = -c^2 \dot{t}^2.$$

Solve for $\dot{t}$:

$$\dot{t} \rightarrow 1.$$

Substitute into $k = A\dot{t}$:

$$k = A(\infty) \cdot 1 = 1.$$

Hence the conserved combination:

$$k^2 - 1 = 0.$$

Insert into the effective-potential equation with $h = 0$:

$$\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) \Big|_{h=0} = 0.$$

Solve for $\frac{1}{2}\dot{r}^2$:

$$\frac{1}{2}\dot{r}^2 = -V_{\text{eff}}(r) \Big|_{h=0} = \frac{c^2 r_G}{2r} - \frac{c^2 r_Q^2}{2r^2}.$$

Impose the turning-point condition $\dot{r} = 0$ at $r = r_*$:

$$\frac{c^2 r_G}{2r_*} - \frac{c^2 r_Q^2}{2r_*^2} = 0.$$

Divide by $c^2/2$:

$$\frac{r_G}{r_*} - \frac{r_Q^2}{r_*^2} = 0.$$

Multiply through by r_*^2/r_G :

$$\boxed{r_* = r_Q^2/r_G.} \quad (12)$$

At $r = r_*$ the radial acceleration $-V'_{\text{eff}}(r_*)$ is positive (the charge term is repulsive at small r): the particle bounces and accelerates back to infinity.

(d) Free fall from r_+ when $r_Q > r_G$

Impose $\dot{r} = 0$ at r_+ with $h = 0$ in the effective-potential equation:

$$V_{\text{eff}}(r_+) \Big|_{h=0} = \frac{1}{2}c^2(k^2 - 1).$$

Use $V_{\text{eff}}|_{h=0} = \frac{1}{2}c^2(A - 1)$:

$$\frac{1}{2}c^2(A(r_+) - 1) = \frac{1}{2}c^2(k^2 - 1).$$

Solve for k^2 :

$$k^2 = A(r_+).$$

Substitute into $\dot{r}^2 = c^2(k^2 - A(r))$:

$$\dot{r}^2 = c^2(A(r_+) - A(r)).$$

Form the difference using $A = 1 - r_G/r + r_Q^2/r^2$:

$$A(r_+) - A(r) = r_G \left(\frac{1}{r} - \frac{1}{r_+} \right) - r_Q^2 \left(\frac{1}{r^2} - \frac{1}{r_+^2} \right).$$

Combine each pair over a common denominator:

$$A(r_+) - A(r) = \frac{r_G(r_+ - r)}{rr_+} - \frac{r_Q^2(r_+^2 - r^2)}{r^2r_+^2}.$$

Factor $r_+^2 - r^2 = (r_+ - r)(r_+ + r)$ in the second term:

$$A(r_+) - A(r) = \frac{r_G(r_+ - r)}{rr_+} - \frac{r_Q^2(r_+ - r)(r_+ + r)}{r^2r_+^2}.$$

Pull out $(r_+ - r)$ over a common denominator $r^2r_+^2$:

$$A(r_+) - A(r) = \frac{(r_+ - r)[r_Grr_+ - r_Q^2(r_+ + r)]}{r^2r_+^2}.$$

Expand the bracket and collect powers of r :

$$r_Grr_+ - r_Q^2(r_+ + r) = (r_Gr_+ - r_Q^2)r - r_Q^2r_+.$$

Factor out $(r_Gr_+ - r_Q^2)$:

$$r_Grr_+ - r_Q^2(r_+ + r) = (r_Gr_+ - r_Q^2) \left(r - \frac{r_Q^2r_+}{r_Gr_+ - r_Q^2} \right).$$

Define $r_- \equiv r_Q^2r_+/(r_Gr_+ - r_Q^2)$:

$$r_Grr_+ - r_Q^2(r_+ + r) = (r_Gr_+ - r_Q^2)(r - r_-).$$

Use $r_Q^2 = r_Gr_*$ in the prefactor:

$$r_Gr_+ - r_Q^2 = r_Gr_+ - r_Gr_* = r_G(r_+ - r_*).$$

Substitute into r_- :

$$r_- = \frac{r_G r_* r_+}{r_G(r_+ - r_*)} = \frac{r_+ r_*}{r_+ - r_*}.$$

Assemble $A(r_+) - A(r)$:

$$A(r_+) - A(r) = \frac{r_G(r_+ - r_*)(r_+ - r)(r - r_-)}{r^2 r_+^2}.$$

Multiply by c^2 :

$$\dot{r}^2 = \frac{c^2 r_G(r_+ - r_*)}{r_+^2 r^2} (r_+ - r)(r - r_-). \quad (13)$$

Both turning points are real with $r_- < r_+$. The particle oscillates between r_+ and r_- , never reaching $r = 0$: the charge term provides an inner centrifugal-like barrier. In Schwarzschild ($Q = 0$) one has $r_* = 0$, $r_- = 0$, so the inner barrier disappears and the particle reaches the singularity.

(e) Proper time $r_+ \rightarrow r_-$

Take the negative square root of (??) for infall ($\dot{r} < 0$):

$$\dot{r} = -\frac{c}{r_+ r} \sqrt{r_G(r_+ - r_*)} \sqrt{(r_+ - r)(r - r_-)}.$$

Rearrange $d\tau = dr/\dot{r}$:

$$d\tau = -\frac{r_+ r}{c \sqrt{r_G(r_+ - r_*)}} \frac{dr}{\sqrt{(r_+ - r)(r - r_-)}}.$$

Integrate from r_+ down to r_- (the minus sign flips the limits):

$$\tau = \frac{r_+}{c \sqrt{r_G(r_+ - r_*)}} \int_{r_-}^{r_+} \frac{r dr}{\sqrt{(r_+ - r)(r - r_-)}}.$$

Substitute $r = \frac{r_+ + r_-}{2} + \frac{r_+ - r_-}{2} u$, $dr = \frac{r_+ - r_-}{2} du$:

$$r_+ - r = \frac{r_+ - r_-}{2}(1 - u), \quad r - r_- = \frac{r_+ - r_-}{2}(1 + u).$$

Multiply the two factors:

$$(r_+ - r)(r - r_-) = \frac{(r_+ - r_-)^2}{4}(1 - u^2).$$

Take the square root:

$$\sqrt{(r_+ - r)(r - r_-)} = \frac{r_+ - r_-}{2} \sqrt{1 - u^2}.$$

Reduce the differential ratio:

$$\frac{dr}{\sqrt{(r_+ - r)(r - r_-)}} = \frac{du}{\sqrt{1 - u^2}}.$$

Replace the integrand:

$$\frac{r dr}{\sqrt{(r_+ - r)(r - r_-)}} = \frac{r}{\sqrt{1 - u^2}} du.$$

Split r into a constant plus an odd u -piece:

$$\int_{-1}^1 \frac{r}{\sqrt{1 - u^2}} du = \frac{r_+ + r_-}{2} \int_{-1}^1 \frac{du}{\sqrt{1 - u^2}} + \frac{r_+ - r_-}{2} \int_{-1}^1 \frac{u du}{\sqrt{1 - u^2}}.$$

The second integral vanishes by parity:

$$\int_{-1}^1 \frac{r}{\sqrt{1-u^2}} du = \frac{r_+ + r_-}{2} \cdot \pi = \frac{\pi(r_+ + r_-)}{2}.$$

Insert into the expression for τ :

$$\tau = \frac{\pi r_+(r_+ + r_-)}{2c\sqrt{r_G(r_+ - r_*)}}.$$

Multiply both sides of $r_- = r_+ r_*/(r_+ - r_*)$ by $(r_+ - r_*)$:

$$r_-(r_+ - r_*) = r_+ r_*.$$

Expand the LHS and rearrange:

$$r_+ r_- = r_*(r_+ + r_-).$$

Solve for r_* :

$$r_* = \frac{r_+ r_-}{r_+ + r_-}.$$

Form $r_+ - r_*$ over a common denominator:

$$r_+ - r_* = \frac{r_+(r_+ + r_-) - r_+ r_-}{r_+ + r_-} = \frac{r_+^2}{r_+ + r_-}.$$

Multiply by r_G and take the square root:

$$\sqrt{r_G(r_+ - r_*)} = r_+ \sqrt{\frac{r_G}{r_+ + r_-}}.$$

Divide both sides into r_+ :

$$\frac{r_+}{\sqrt{r_G(r_+ - r_*)}} = \sqrt{\frac{r_+ + r_-}{r_G}}.$$

Insert this back into τ :

$$\tau = \frac{\pi(r_+ + r_-)}{2c} \sqrt{\frac{r_+ + r_-}{r_G}}.$$

Combine the powers of $(r_+ + r_-)$:

$$\tau = \frac{\pi(r_+ + r_-)^{3/2}}{2c\sqrt{r_G}}.$$

Multiply and divide by r_G to obtain the paper's form:

$$\boxed{\tau = \frac{\pi r_G}{2c} \left(\frac{r_+ + r_-}{r_G} \right)^{3/2}}. \quad (14)$$

Question 3 — Anisotropic cosmology

Problem. Bianchi-I metric $ds^2 = -c^2 dt^2 + e^{2\alpha} dx^2 + e^{2\beta} (dy^2 + dz^2)$ with anisotropic perfect fluid (pressures $w_A \rho c^2$ along x , $w_B \rho c^2$ along y, z). (a) Christoffels. (b) $T^{\mu\nu}_{;\mu} = 0$ gives $\rho \propto e^{-(1+w_A)\alpha - 2(1+w_B)\beta}$. (c) Einstein equations. (d) $w_A = w_B = -1$: isotropic stable de Sitter. (e) Volume growth. (f) Two observational tests of cosmic anisotropy.

(a) Christoffel symbols

Set $x^0 = ct$, so $g_{00} = -1$ and $\partial_0 = (1/c)\partial_t$. Use the diagonal Christoffel formula:

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\sigma}(g_{\sigma\mu,\nu} + g_{\sigma\nu,\mu} - g_{\mu\nu,\sigma}).$$

Compute Γ_{xx}^0 ($\sigma = \lambda = 0$, $g_{0x} = 0$ so only the last term survives):

$$\Gamma_{xx}^0 = -\frac{1}{2}g^{00}g_{xx,0} = -\frac{1}{2}(-1)\partial_0 e^{2\alpha} = \frac{1}{2}\partial_0 e^{2\alpha}.$$

Substitute $\partial_0 e^{2\alpha} = (2\dot{\alpha}/c)e^{2\alpha}$:

$$\Gamma_{xx}^0 = \frac{\dot{\alpha} e^{2\alpha}}{c}.$$

Repeat with $g_{yy} = g_{zz} = e^{2\beta}$:

$$\Gamma_{yy}^0 = \Gamma_{zz}^0 = \frac{\dot{\beta} e^{2\beta}}{c}.$$

Compute Γ_{0x}^x ($\sigma = \lambda = x$; only $g_{xx,0}$ contributes):

$$\Gamma_{0x}^x = \frac{1}{2}g^{xx}g_{xx,0} = \frac{1}{2}e^{-2\alpha}\partial_0 e^{2\alpha}.$$

Cancel the exponential:

$$\Gamma_{0x}^x = \frac{\dot{\alpha}}{c}.$$

Repeat for y, z :

$$\Gamma_{0y}^y = \Gamma_{0z}^z = \frac{\dot{\beta}}{c}.$$

$$\Gamma_{xx}^0 = \frac{\dot{\alpha}}{c}e^{2\alpha}, \quad \Gamma_{yy}^0 = \Gamma_{zz}^0 = \frac{\dot{\beta}}{c}e^{2\beta}, \quad \Gamma_{0x}^x = \frac{\dot{\alpha}}{c}, \quad \Gamma_{0y}^y = \Gamma_{0z}^z = \frac{\dot{\beta}}{c}.$$

(15)

(b) Continuity for ρ

Raise the second index of $T^\mu{}_\nu$ with $g^{\nu\nu}$:

$$T^{00} = g^{00}T^0{}_0 = (-1)(-\rho c^2) = \rho c^2.$$

Similarly for the spatial components:

$$T^{xx} = e^{-2\alpha}w_A\rho c^2, \quad T^{yy} = T^{zz} = e^{-2\beta}w_B\rho c^2.$$

Expand the conservation $\nabla_\mu T^{\mu 0} = 0$ in coordinates, using $T^{i0} = 0$:

$$\partial_0 T^{00} + \Gamma_{\mu 0}^\mu T^{00} + \Gamma_{\mu\lambda}^0 T^{\mu\lambda} = 0.$$

Differentiate T^{00} in x^0 :

$$\partial_0 T^{00} = \frac{1}{c}\partial_t(\rho c^2) = c\dot{\rho}.$$

Sum the contracted Christoffel:

$$\Gamma_{\mu 0}^\mu = \Gamma_{x0}^x + \Gamma_{y0}^y + \Gamma_{z0}^z = \frac{\dot{\alpha} + 2\dot{\beta}}{c}.$$

Insert these two results:

$$c\dot{\rho} + \frac{\dot{\alpha} + 2\dot{\beta}}{c}\rho c^2 + \Gamma_{\mu\lambda}^0 T^{\mu\lambda} = 0.$$

Evaluate the source term using $\Gamma_{\mu\lambda}^0 T^{\mu\lambda} = \Gamma_{xx}^0 T^{xx} + 2\Gamma_{yy}^0 T^{yy}$:

$$\Gamma_{\mu\lambda}^0 T^{\mu\lambda} = \frac{\dot{\alpha} e^{2\alpha}}{c} \cdot e^{-2\alpha} w_A \rho c^2 + 2 \cdot \frac{\dot{\beta} e^{2\beta}}{c} \cdot e^{-2\beta} w_B \rho c^2.$$

Cancel the exponentials:

$$\Gamma_{\mu\lambda}^0 T^{\mu\lambda} = \frac{\dot{\alpha} w_A + 2\dot{\beta} w_B}{c} \rho c^2.$$

Substitute back and divide by $c\rho$:

$$\dot{\rho}/\rho + (\dot{\alpha} + 2\dot{\beta}) + (\dot{\alpha} w_A + 2\dot{\beta} w_B) = 0.$$

Collect $\dot{\alpha}$ and $\dot{\beta}$:

$$\frac{\dot{\rho}}{\rho} = -(1 + w_A)\dot{\alpha} - 2(1 + w_B)\dot{\beta}.$$

Integrate with respect to t :

$$\ln \rho = -(1 + w_A)\alpha(t) - 2(1 + w_B)\beta(t) + \text{const.}$$

Exponentiate:

$$\boxed{\rho(t) \propto e^{-(1+w_A)\alpha(t)-2(1+w_B)\beta(t)}}. \quad (16)$$

(c) Einstein field equations

Take the determinant of the diagonal metric:

$$|g| = e^{2\alpha} \cdot e^{2\beta} \cdot e^{2\beta} = e^{2\alpha+4\beta}, \quad \ln |g| = 2\alpha + 4\beta.$$

Substitute into the given Ricci formula:

$$R_{\mu\nu} = \frac{1}{2} \partial_\mu \partial_\nu \ln |g| - \partial_\gamma \Gamma_{\mu\nu}^\gamma + \Gamma_{\mu\gamma}^\delta \Gamma_{\nu\delta}^\gamma - \frac{1}{2} \Gamma_{\mu\nu}^\gamma \partial_\gamma \ln |g|.$$

R_{00} . Compute the second derivative of $\ln |g|$:

$$\frac{1}{2} \partial_0 \partial_0 \ln |g| = \frac{1}{2c^2} \partial_t^2 (2\alpha + 4\beta) = \frac{1}{c^2} (\ddot{\alpha} + 2\ddot{\beta}).$$

Drop the middle terms since $\Gamma_{00}^\gamma = 0$. Sum the quadratic Christoffel term:

$$\Gamma_{0\gamma}^\delta \Gamma_{0\delta}^\gamma = (\Gamma_{0x}^x)^2 + (\Gamma_{0y}^y)^2 + (\Gamma_{0z}^z)^2 = \frac{\dot{\alpha}^2 + 2\dot{\beta}^2}{c^2}.$$

Add the two surviving contributions:

$$R_{00} = \frac{1}{c^2} (\ddot{\alpha} + 2\ddot{\beta}) + \frac{1}{c^2} (\dot{\alpha}^2 + 2\dot{\beta}^2).$$

Combine into one bracket:

$$R_{00} = \frac{1}{c^2} (\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 2\dot{\beta}^2).$$

R_{xx} . The second derivative $\partial_x \partial_x \ln |g| = 0$ since α, β depend only on t . The only nonzero Γ_{xx}^γ is Γ_{xx}^0 ; compute its derivative:

$$\partial_0 \Gamma_{xx}^0 = \frac{1}{c^2} \partial_t (\dot{\alpha} e^{2\alpha}).$$

Apply the product rule:

$$\partial_0 \Gamma_{xx}^0 = \frac{e^{2\alpha}}{c^2} (\ddot{\alpha} + 2\dot{\alpha}^2).$$

Sum the quadratic Christoffel term using the pairs $(\delta, \gamma) = (0, x)$ and $(x, 0)$:

$$\Gamma_{x\gamma}^{\delta} \Gamma_{x\delta}^{\gamma} = \Gamma_{xx}^0 \Gamma_{x0}^x + \Gamma_{x0}^x \Gamma_{xx}^0 = 2 \cdot \frac{e^{2\alpha} \dot{\alpha}}{c} \cdot \frac{\dot{\alpha}}{c} = \frac{2e^{2\alpha} \dot{\alpha}^2}{c^2}.$$

Evaluate $\partial_0 \ln |g|$:

$$\partial_0 \ln |g| = \frac{1}{c} (2\dot{\alpha} + 4\dot{\beta}).$$

Form the last term:

$$\frac{1}{2} \Gamma_{xx}^0 \partial_0 \ln |g| = \frac{1}{2} \cdot \frac{e^{2\alpha} \dot{\alpha}}{c} \cdot \frac{2\dot{\alpha} + 4\dot{\beta}}{c} = \frac{e^{2\alpha} \dot{\alpha} (\dot{\alpha} + 2\dot{\beta})}{c^2}.$$

Assemble $R_{xx} = -\partial_0 \Gamma_{xx}^0 + \Gamma_{x\gamma}^{\delta} \Gamma_{x\delta}^{\gamma} - \frac{1}{2} \Gamma_{xx}^0 \partial_0 \ln |g|$:

$$R_{xx} = \frac{e^{2\alpha}}{c^2} [-(\ddot{\alpha} + 2\dot{\alpha}^2) + 2\dot{\alpha}^2 - \dot{\alpha}(\dot{\alpha} + 2\dot{\beta})].$$

Cancel the $2\dot{\alpha}^2$ terms inside the bracket:

$$R_{xx} = \frac{e^{2\alpha}}{c^2} [-\ddot{\alpha} - \dot{\alpha}(\dot{\alpha} + 2\dot{\beta})].$$

Expand the second piece:

$$R_{xx} = -\frac{e^{2\alpha}}{c^2} (\ddot{\alpha} + \dot{\alpha}^2 + 2\dot{\alpha}\dot{\beta}).$$

R_{yy} (= R_{zz} by symmetry). Replay the same calculation with $\Gamma_{yy}^0 = (\dot{\beta}/c)e^{2\beta}$ and $\Gamma_{y0}^y = \dot{\beta}/c$:

$$-\partial_0 \Gamma_{yy}^0 = -\frac{e^{2\beta}}{c^2} (\ddot{\beta} + 2\dot{\beta}^2), \quad \Gamma_{y\gamma}^{\delta} \Gamma_{y\delta}^{\gamma} = \frac{2e^{2\beta} \dot{\beta}^2}{c^2}.$$

Form the trace term:

$$\frac{1}{2} \Gamma_{yy}^0 \partial_0 \ln |g| = \frac{e^{2\beta} \dot{\beta} (\dot{\alpha} + 2\dot{\beta})}{c^2}.$$

Combine and cancel $2\dot{\beta}^2$ inside the bracket:

$$R_{yy} = -\frac{e^{2\beta}}{c^2} (\ddot{\beta} + \dot{\beta}^2 + 2\dot{\beta}\dot{\alpha}).$$

Ricci scalar. Contract with $g^{\mu\nu}$:

$$R = g^{00} R_{00} + g^{xx} R_{xx} + g^{yy} R_{yy} + g^{zz} R_{zz} = -R_{00} + e^{-2\alpha} R_{xx} + 2e^{-2\beta} R_{yy}.$$

Substitute the components:

$$R = -\frac{1}{c^2} (\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 2\dot{\beta}^2) - \frac{1}{c^2} (\ddot{\alpha} + \dot{\alpha}^2 + 2\dot{\alpha}\dot{\beta}) - \frac{2}{c^2} (\ddot{\beta} + \dot{\beta}^2 + 2\dot{\beta}\dot{\alpha}).$$

Collect like terms (the coefficient of $\dot{\beta}^2$ is $2 + 0 + 4 = 6$, halved gives $3\dot{\beta}^2$ after factoring $2/c^2$):

$$R = -\frac{2}{c^2} (\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 3\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}).$$

Einstein equations. The Einstein equation reads $R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = -\frac{8\pi G}{c^4} T_{\mu\nu}$.
 tt -component, $g_{00} = -1$, $T_{00} = \rho c^2$:

$$G_{00} = R_{00} + \frac{1}{2} R.$$

Substitute the components and add:

$$G_{00} = \frac{1}{c^2}(\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 2\dot{\beta}^2) - \frac{1}{c^2}(\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 3\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}).$$

Cancel the $\ddot{\alpha}, \ddot{\beta}, \dot{\alpha}^2$ terms:

$$G_{00} = -\frac{1}{c^2}(\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}).$$

Set $G_{00} = -\frac{8\pi G}{c^4}\rho c^2 = -\frac{8\pi G}{c^2}\rho$:

$$-\frac{1}{c^2}(\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}) = -\frac{8\pi G}{c^2}\rho.$$

Multiply through by $-c^2$:

$$\boxed{\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta} = 8\pi G\rho.} \quad (17)$$

xx -component, $g_{xx} = e^{2\alpha}$, $T_{xx} = g_{xx}^2 T^{xx} = e^{2\alpha} w_A \rho c^2$:

$$G_{xx} = R_{xx} - \frac{1}{2}R e^{2\alpha}.$$

Insert components:

$$G_{xx} = -\frac{e^{2\alpha}}{c^2}(\ddot{\alpha} + \dot{\alpha}^2 + 2\dot{\alpha}\dot{\beta}) + \frac{e^{2\alpha}}{c^2}(\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 3\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}).$$

Cancel $\ddot{\alpha}, \dot{\alpha}^2, 2\dot{\alpha}\dot{\beta}$:

$$G_{xx} = \frac{e^{2\alpha}}{c^2}(2\ddot{\beta} + 3\dot{\beta}^2).$$

Set $G_{xx} = -\frac{8\pi G}{c^4}e^{2\alpha}w_A \rho c^2$:

$$\frac{e^{2\alpha}}{c^2}(2\ddot{\beta} + 3\dot{\beta}^2) = -\frac{8\pi G}{c^2}e^{2\alpha}w_A \rho.$$

Cancel $e^{2\alpha}/c^2$:

$$\boxed{2\ddot{\beta} + 3\dot{\beta}^2 = -8\pi G w_A \rho.} \quad (18)$$

yy -component, $T_{yy} = e^{2\beta} w_B \rho c^2$:

$$G_{yy} = R_{yy} - \frac{1}{2}R e^{2\beta}.$$

Insert components:

$$G_{yy} = -\frac{e^{2\beta}}{c^2}(\ddot{\beta} + \dot{\alpha}\dot{\beta} + 2\dot{\beta}^2) + \frac{e^{2\beta}}{c^2}(\ddot{\alpha} + 2\ddot{\beta} + \dot{\alpha}^2 + 3\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}).$$

Combine like terms:

$$G_{yy} = \frac{e^{2\beta}}{c^2}(\ddot{\alpha} + \ddot{\beta} + \dot{\alpha}^2 + \dot{\beta}^2 + \dot{\alpha}\dot{\beta}).$$

Set $G_{yy} = -\frac{8\pi G}{c^2}e^{2\beta}w_B \rho$ and cancel $e^{2\beta}/c^2$:

$$\boxed{\ddot{\alpha} + \ddot{\beta} + \dot{\alpha}^2 + \dot{\beta}^2 + \dot{\alpha}\dot{\beta} = -8\pi G w_B \rho.} \quad (19)$$

(d) **Cosmological-constant case** $w_A = w_B = -1$

Substitute $w_A = w_B = -1$ into the exponent of (??):

$$-(1 + w_A)\alpha - 2(1 + w_B)\beta = 0.$$

Exponentiate:

$$\rho = \rho_0 = \text{const.}$$

Impose the fixed-point condition $\ddot{\alpha} = \ddot{\beta} = 0$ in (??):

$$3\dot{\beta}^2 = 8\pi G\rho_0.$$

Subtract (??) from this result:

$$3\dot{\beta}^2 - (\dot{\beta}^2 + 2\dot{\alpha}\dot{\beta}) = 0.$$

Simplify the LHS:

$$2\dot{\beta}(\dot{\beta} - \dot{\alpha}) = 0.$$

Take the non-trivial root $\dot{\beta} \neq 0$:

$$\dot{\alpha} = \dot{\beta}.$$

Hence the expansion is isotropic. Solve $3\dot{\beta}^2 = 8\pi G\rho_0$ for the common rate q :

$$\dot{\alpha} = \dot{\beta} = \sqrt{\frac{8\pi G\rho_0}{3}} \equiv q.$$

Check (??): $0 + 0 + q^2 + q^2 + q^2 = 3q^2 = 8\pi G\rho_0$, consistent. The fixed-point solution is exponential:

$$\alpha = \beta = qt.$$

Linear stability. Set $\alpha = qt + \delta\alpha$, $\beta = qt + \delta\beta$ with $\rho = \rho_0$ unchanged. Linearise the LHS of (??):

$$\dot{\beta}^2 \approx q^2 + 2q\dot{\delta}\beta, \quad 2\dot{\alpha}\dot{\beta} \approx 2q^2 + 2q\dot{\delta}\alpha + 2q\dot{\delta}\beta.$$

Add and subtract the zeroth-order $3q^2 = 8\pi G\rho_0$:

$$2q\dot{\delta}\beta + 2q\dot{\delta}\alpha + 2q\dot{\delta}\beta = 0.$$

Combine like terms:

$$2q(\dot{\delta}\alpha + 2\dot{\delta}\beta) = 0.$$

Divide by $2q$:

$$\dot{\delta}\alpha + 2\dot{\delta}\beta = 0.$$

Linearise (??):

$$2\ddot{\delta}\beta + 3(q^2 + 2q\dot{\delta}\beta) = 3q^2.$$

Cancel the zeroth-order $3q^2$:

$$2\ddot{\delta}\beta + 6q\dot{\delta}\beta = 0.$$

Divide by 2:

$$\ddot{\delta}\beta + 3q\dot{\delta}\beta = 0.$$

Set $u = \dot{\delta}\beta$ and solve $\dot{u} = -3qu$:

$$\dot{\delta}\beta \propto e^{-3qt} \xrightarrow{t \rightarrow \infty} 0.$$

Use the first linearisation:

$$\dot{\delta}\alpha = -2\dot{\delta}\beta \propto e^{-3qt} \xrightarrow{t \rightarrow \infty} 0.$$

Both shear modes decay; the isotropic de Sitter solution is stable.

$\rho = \text{const}, \quad \alpha = \beta = qt, \quad \text{shear} \propto e^{-3qt}.$
--

(e) Physical volume of a comoving region

The induced 3-metric on a constant- t slice has determinant $|g_3| = e^{2\alpha}e^{2\beta}e^{2\beta} = e^{2\alpha+4\beta}$. Volume element:

$$dV = \sqrt{|g_3|} dx dy dz = e^{\alpha+2\beta} dx dy dz.$$

Insert $\alpha = \beta = qt$ from (d):

$$dV = e^{qt+2qt} dx dy dz = e^{3qt} dx dy dz.$$

Integrate over the fixed comoving volume V_0 :

$$V(t) = V_0 e^{3qt}.$$

$$\boxed{V(t) \propto e^{3qt}.} \quad (20)$$

(f) Detecting cosmic anisotropy

Two probes are:

(i) CMB anisotropy pattern. Bianchi-I expansion imprints a direction-dependent quadrupole / octopole in the CMB temperature map (different recession rates in x vs y, z produce a preferred-axis signature). Stack with foreground-cleaned all-sky maps (Planck) and constrain shear $\Sigma \equiv (\dot{\alpha} - \dot{\beta})/H$ from the absence of an excess large-angle quadrupole.

(ii) Direction-dependent Hubble diagram. Compile Type Ia supernovae (or BAO standard rulers, or strong-lensing time delays) across the sky and fit $H_0(\hat{n})$ as a function of direction. A non-zero dipole / quadrupole in $H_0(\hat{n})$ above the peculiar-velocity floor implies anisotropic expansion.

Question 4 — Gravitational waves from a radial binary

Problem. Two equal masses μ on the z -axis, separation $\Delta z(t) = L + a \cos \omega t$. (a) Identify configuration; compute $I^{ij} = \int T^{00} r^i r^j d^3r$. (b) Linearised gravity, find $\bar{h}_{ij}^{TT}$ propagating along $\hat{x}$. (c) Motion of a small ring of test masses centred on, perpendicular to, $\hat{x}$. (d) Total GW luminosity. (e) Energy radiated per period; discuss L/a dependence.

(a) Physical configuration and quadrupole moment

A pair of equal masses on a spring along $\hat{z}$ (or two stars in a radial-only oscillation about a fixed mean separation L , amplitude a). Place the centre of mass at the origin so $z_1 + z_2 = 0$ and $z_2 - z_1 = \Delta z$:

$$z_{1,2}(t) = \mp \frac{1}{2} (L + a \cos \omega t), \quad x_{1,2} = y_{1,2} = 0.$$

For point particles, the energy density is a sum of delta functions:

$$T^{00}(t, \vec{r}) = \mu c^2 [\delta^3(\vec{r} - \vec{r}_1) + \delta^3(\vec{r} - \vec{r}_2)].$$

Insert into the integral and use the sifting property:

$$I^{ij} = \mu c^2 (r_1^i r_1^j + r_2^i r_2^j).$$

Since $x_n = y_n = 0$, only the (z, z) component is non-zero:

$$I^{zz} = \mu c^2 (z_1^2 + z_2^2).$$

Substitute $z_1^2 = z_2^2 = (L + a \cos \omega t)^2/4$:

$$I^{zz} = \mu c^2 \cdot 2 \cdot \frac{(L + a \cos \omega t)^2}{4}.$$

Simplify the prefactor:

$$I^{ij} = \delta_z^i \delta_z^j \frac{\mu c^2}{2} (L + a \cos \omega t)^2. \quad (21)$$

(b) Transverse-traceless metric perturbation along $\hat{x}$

Linearised quadrupole formula in Lorenz gauge (mass-density quadrupole $J^{ij} = I^{ij}/c^2$):

$$\bar{h}^{ij}(t, \vec{r}) = \frac{2G}{c^4 r} \ddot{J}^{ij}(t - r/c) = \frac{2G}{c^6 r} \ddot{I}^{ij}(t - r/c). \quad (22)$$

Differentiate $I^{zz} = \frac{\mu c^2}{2} (L + a \cos \omega t)^2$ once by the chain rule:

$$\dot{I}^{zz} = \frac{\mu c^2}{2} \cdot 2(L + a \cos \omega t)(-a\omega \sin \omega t).$$

Simplify the prefactor:

$$\dot{I}^{zz} = -\mu c^2 a \omega \sin \omega t (L + a \cos \omega t).$$

Differentiate again by the product rule:

$$\ddot{I}^{zz} = -\mu c^2 a \omega [\omega \cos \omega t (L + a \cos \omega t) + \sin \omega t (-a\omega \sin \omega t)].$$

Expand:

$$\ddot{I}^{zz} = -\mu c^2 a \omega^2 [L \cos \omega t + a \cos^2 \omega t - a \sin^2 \omega t].$$

Use $\cos^2 \omega t - \sin^2 \omega t = \cos 2\omega t$:

$$\ddot{I}^{zz} = -\mu c^2 \omega^2 [La \cos \omega t + a^2 \cos 2\omega t].$$

Project to TT along $\hat{n} = \hat{x}$. Then $P_{ij} = \delta_{ij} - n_i n_j = \text{diag}(0, 1, 1)$ in (x, y, z) , and for a tensor with only A_{zz} nonzero, $P_{ab} A^{ab} = A_{zz}$:

$$A_{ij}^{TT} = P_{ia} P_{jb} A^{ab} - \frac{1}{2} P_{ij} P_{ab} A^{ab} = P_{iz} P_{jz} A_{zz} - \frac{1}{2} P_{ij} A_{zz}.$$

Evaluate the (z, z) component:

$$A_{zz}^{TT} = P_{zz}^2 A_{zz} - \frac{1}{2} P_{zz} A_{zz} = A_{zz} - \frac{1}{2} A_{zz} = \frac{1}{2} A_{zz}.$$

Evaluate the (y, y) component:

$$A_{yy}^{TT} = P_{yz}^2 A_{zz} - \frac{1}{2} P_{yy} A_{zz} = 0 - \frac{1}{2} A_{zz} = -\frac{1}{2} A_{zz}.$$

Evaluate the off-diagonal:

$$A_{yz}^{TT} = P_{yz} P_{zz} A_{zz} - \frac{1}{2} P_{yz} A_{zz} = 0.$$

Apply (??) with $u \equiv t - r/c$, taking $A = \ddot{I}$:

$$\bar{h}^{zz}(t, \vec{r}) = \frac{2G}{c^6 r} \cdot (-\mu c^2 \omega^2) [La \cos \omega u + a^2 \cos 2\omega u].$$

Simplify:

$$\bar{h}^{zz} = -\frac{2G\mu\omega^2}{c^4 r} [La \cos \omega u + a^2 \cos 2\omega u].$$

Multiply by the TT projector $\frac{1}{2}$ on the (z, z) piece:

$$\bar{h}_{zz}^{TT} = -\bar{h}_{yy}^{TT} = -\frac{G\mu\omega^2}{c^4 r} [La \cos \omega u + a^2 \cos 2\omega u]. \quad (23)$$

(c) Motion of the test-mass ring

The wave is +-polarised in the (y, z) plane. Apply the geodesic-deviation equation:

$$\ddot{\xi}^i = \frac{1}{2} \ddot{h}_{ij}^{TT} \xi^j.$$

Integrate twice with ξ_0^j constant:

$$\xi^i(t) = \xi_0^i + \frac{1}{2} \bar{h}_{ij}^{TT}(t) \xi_0^j.$$

Parametrise the initial ring of radius δR in the (y, z) plane:

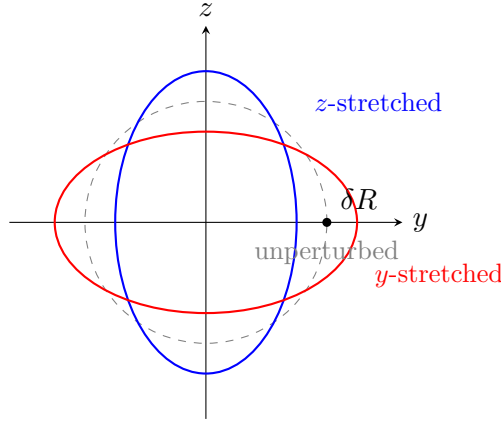
$$\xi_0^y = \delta R \cos \varphi, \quad \xi_0^z = \delta R \sin \varphi.$$

Apply $\bar{h}_{yy}^{TT} = -\bar{h}_{zz}^{TT}$ to each:

$$\delta y(t) = \delta R \cos \varphi \left[1 + \frac{1}{2} \bar{h}_{yy}^{TT}(t) \right] = \delta R \cos \varphi \left[1 - \frac{1}{2} \bar{h}_{zz}^{TT}(t) \right].$$

$$\delta z(t) = \delta R \sin \varphi \left[1 + \frac{1}{2} \bar{h}_{zz}^{TT}(t) \right].$$

The circle deforms into an ellipse whose principal axes coincide with $\hat{y}, \hat{z}$, with strain alternating at the two harmonics ω (dominant when $L \gg a$) and 2ω . The ring breathes between a z -stretched / y -squeezed ellipse and a y -stretched / z -squeezed ellipse, with constant area to linear order in $\bar{h}$.



(d) Total gravitational-wave luminosity

Integrate $\mathcal{F}_r$ over the sphere $4\pi r^2$ to obtain the standard Einstein quadrupole luminosity in terms of the trace-free reduced mass-quadrupole $\bar{Q}_{ij} = Q_{ij} - \frac{1}{3} Q^k_k \delta_{ij}$ with $Q_{ij} \equiv I_{ij}/c^2$:

$$\mathcal{L} = \frac{G}{5c^5} \langle \ddot{\bar{Q}}_{ij} \ddot{\bar{Q}}^{ij} \rangle. \quad (24)$$

Read off $Q_{zz} = \mu(L + a \cos \omega t)^2/2$ with all other $Q_{ij} = 0$. Take the trace:

$$Q^k_k = Q_{zz}.$$

Form the trace-free components:

$$\bar{Q}_{zz} = Q_{zz} - \frac{1}{3} Q_{zz} = \frac{2}{3} Q_{zz}, \quad \bar{Q}_{xx} = \bar{Q}_{yy} = -\frac{1}{3} Q_{zz}.$$

Expand Q_{zz} using $\cos^2 \omega t = \frac{1}{2}(1 + \cos 2\omega t)$:

$$Q_{zz} = \frac{\mu}{2} [L^2 + 2La \cos \omega t + a^2 \cos^2 \omega t].$$

Apply the identity:

$$Q_{zz} = \frac{\mu}{2} \left[L^2 + \frac{a^2}{2} + 2La \cos \omega t + \frac{a^2}{2} \cos 2\omega t \right].$$

Differentiate three times in t (constants drop; $\cos n\omega t \xrightarrow{(d/dt)^3} n^3 \omega^3 \sin n\omega t$):

$$\ddot{Q}_{zz} = \frac{\mu}{2} [2La\omega^3 \sin \omega t + \frac{a^2}{2} \cdot 8\omega^3 \sin 2\omega t].$$

Simplify the coefficients:

$$\ddot{Q}_{zz} = \mu [La\omega^3 \sin \omega t + 2a^2\omega^3 \sin 2\omega t] \equiv \mu S(t).$$

Square the source:

$$S^2 = L^2 a^2 \omega^6 \sin^2 \omega t + 4a^4 \omega^6 \sin^2 2\omega t + 4La^3 \omega^6 \sin \omega t \sin 2\omega t.$$

Apply $\langle \sin^2 \omega t \rangle = \langle \sin^2 2\omega t \rangle = \frac{1}{2}$ and the orthogonality of $\sin \omega t, \sin 2\omega t$:

$$\langle S^2 \rangle = \frac{1}{2} L^2 a^2 \omega^6 + 2a^4 \omega^6.$$

Factor $\frac{1}{2} a^2 \omega^6$:

$$\langle S^2 \rangle = \frac{1}{2} a^2 \omega^6 (L^2 + 4a^2).$$

Sum the trace-free squared components, using $\ddot{Q}_{zz} = (2/3)\mu S$, $\ddot{Q}_{xx} = \ddot{Q}_{yy} = -(1/3)\mu S$:

$$\ddot{Q}_{ij} \ddot{Q}^{ij} = \left(\frac{4}{9} + 2 \cdot \frac{1}{9} \right) \mu^2 S^2.$$

Combine the fractions:

$$\ddot{Q}_{ij} \ddot{Q}^{ij} = \frac{2}{3} \mu^2 S^2.$$

Average:

$$\langle \ddot{Q}_{ij} \ddot{Q}^{ij} \rangle = \frac{2}{3} \mu^2 \cdot \frac{1}{2} a^2 \omega^6 (L^2 + 4a^2) = \frac{\mu^2 a^2 \omega^6 (L^2 + 4a^2)}{3}.$$

Insert into (??):

$$\mathcal{L} = \frac{G}{5c^5} \cdot \frac{\mu^2 a^2 \omega^6 (L^2 + 4a^2)}{3}.$$

Multiply the constants:

$$\boxed{\mathcal{L} = \frac{G \mu^2 \omega^6 a^2 (L^2 + 4a^2)}{15 c^5}}. \quad (25)$$

(e) Energy radiated per period

Multiply the luminosity by the period $T = 2\pi/\omega$:

$$\Delta E = \mathcal{L} T = \frac{G \mu^2 \omega^6 a^2 (L^2 + 4a^2)}{15 c^5} \cdot \frac{2\pi}{\omega}.$$

Simplify the ω power:

$$\Delta E = \frac{2\pi G \mu^2 \omega^5 a^2 (L^2 + 4a^2)}{15 c^5}.$$

Compute the maximum speed of each mass from $\dot{z}_{1,2} = \pm \frac{1}{2} a \omega \sin \omega t$:

$$v_{\max} = \frac{a\omega}{2} \implies a^2 \omega^2 = 4v_{\max}^2.$$

Define a separation-scale velocity:

$$v_L \equiv \frac{L\omega}{2} \implies L^2\omega^2 = 4v_L^2.$$

Rewrite $\omega^5 a^2$ using the first relation:

$$\omega^5 a^2 = \omega^3 \cdot a^2 \omega^2 = 4\omega^3 v_{\max}^2.$$

Substitute:

$$\Delta E = \frac{2\pi G\mu^2}{15c^5} \cdot 4\omega^3 v_{\max}^2 (L^2 + 4a^2) = \frac{8\pi G\mu^2 \omega^3 v_{\max}^2 (L^2 + 4a^2)}{15c^5}.$$

Replace $L^2 = 4v_L^2/\omega^2$ and $4a^2 = 16v_{\max}^2/\omega^2$:

$$\Delta E = \frac{8\pi G\mu^2 \omega^3 v_{\max}^2}{15c^5} \cdot \frac{4v_L^2 + 16v_{\max}^2}{\omega^2}.$$

Collect the ω factors:

$$\Delta E = \frac{8\pi G\mu^2 \omega v_{\max}^2 (4v_L^2 + 16v_{\max}^2)}{15c^5}.$$

Factor a 4:

$$\boxed{\Delta E = \frac{32\pi G\mu^2 \omega v_{\max}^2 (v_L^2 + 4v_{\max}^2)}{15c^5}} \quad (26)$$

Discussion of L/a .

- $L/a \gg 1$ (small radial oscillation about a large separation): the ω -harmonic dominates, $\Delta E \approx \frac{2\pi G\mu^2 \omega^5 L^2 a^2}{15c^5} = \frac{32\pi G\mu^2 \omega v_L^2 v_{\max}^2}{15c^5}$. Energy loss grows as $(L/a)^2$ at fixed a , so the radiation is sourced by the mass-monopole modulation about a large equilibrium length.
- $L/a \rightarrow 0$ (pure breathing through the centre): only the 2ω harmonic survives, $\Delta E \rightarrow \frac{8\pi G\mu^2 \omega^5 a^4}{15c^5} = \frac{128\pi G\mu^2 \omega v_{\max}^4}{15c^5}$. Independent of L ; this is the lowest-order quadrupole emission of a symmetric oscillator.
- Crossover at $L/a \sim 2$: the two contributions are comparable, and the GW signal contains a clean two-tone spectrum at ω and 2ω .